

Learn Cryptocurrency: Beginner's Guide

Lesson 2: How Blockchain Works

How Blockchain Works - Study Notes

Key Concepts

1. **Blocks** – Containers that store a batch of transactions, a timestamp, a reference to the previous block (hash), and a unique block hash.
2. **Decentralization** – A network of many independent nodes that all keep a copy of the ledger, eliminating a single point of control or failure.
3. **Consensus Mechanisms** – Rules that let the distributed nodes agree on which block is the next valid addition to the chain (e.g., Proof of Work, Proof of Stake).
4. **Hash Functions** – Cryptographic algorithms that turn data into a fixed size string; tiny changes in input produce completely different outputs, ensuring integrity.
5. **Immutability & Security** – Once a block is linked to the chain, altering it would require re-computing every subsequent block's hash, which is computationally infeasible in a well-secured network.

Summary

A blockchain is a **distributed ledger** made up of sequentially linked **blocks**. Each block holds a collection of transactions and a cryptographic **hash** of the previous block, forming an immutable chain. Because the ledger is stored on many **decentralized nodes**, no single entity controls the data, and every participant can verify the history independently.

To add a new block, the network must reach **consensus**—a common agreement on which block is valid. Consensus mechanisms such as **Proof of Work (PoW)** require nodes (miners) to solve a computational puzzle, while **Proof of Stake (PoS)** selects validators based on the amount of cryptocurrency they lock up. Once consensus is achieved, the new block is broadcast, each node updates its copy, and the process repeats, creating a continuously growing, tamper-evident record.

Important Points

- **Block Structure**
 - Header: previous block hash, timestamp, nonce (PoW), Merkle root of transactions.
 - Body: list of transactions (or other data).
- **Hash Function Properties**
 - Deterministic, quick to compute, pre-image resistant, avalanche effect.
- **Decentralization Benefits**
 - Reduces censorship, improves fault tolerance, enhances transparency.
- **Consensus Types (simplified)**
 - **Proof of Work:** solves a hash puzzle; security comes from computational effort.
 - **Proof of Stake:** validators stake tokens; security comes from economic value at risk.

- **Other examples:** Delegated PoS, Practical Byzantine Fault Tolerance (PBFT).

- **Immutability**

- Changing a transaction requires re hashing that block **and** all following blocks, which is practically impossible in a large network.

- **Simple Diagram (textual)**

...

[Node A] [Node B] [Node C] ... (all keep a copy)

!“ “ “

+-----+ +-----+ +-----+

| Block0 |' | Block1 |' | Block2 |' ...

+-----+ +-----+ +-----+

^ ^ ^

| | |

Prev hash Prev hash Prev hash

...

Examples

1. **Bitcoin Transaction**

- Alice sends 0.5 /BTC to Bob. The transaction is broadcast to the network.
- Miners collect many such transactions into a candidate block, compute a PoW hash that meets the difficulty target, and propose the block.
- Once a majority of nodes accept the block, it is added after Block /N, becoming Block /N+1. The ledger now permanently records Alice !' Bob.

2. **Supply Chain Tracking (Private PoS Blockchain)**

- A food company records each step: farm !' processor !' distributor !' retailer.
- Each participant adds a block with data (timestamp, location, QR code).
- Validators (selected by PoS) confirm the block, ensuring only authorized parties can write, while all parties can read the immutable history, improving traceability and trust.

Quick Review

1. What two pieces of information link one block to the previous block?
2. How does decentralization improve security compared to a single server database?
3. In Proof of Work, what is the purpose of the “nonce”?
4. Why is it practically impossible to alter a transaction deep inside an existing blockchain?
5. Name one advantage and one disadvantage of Proof of Stake versus Proof of Work.

Further Reading

- **Hash Functions & Merkle Trees** – deeper dive into how transaction data is summarized.
- **Detailed Consensus Algorithms** – PoW, PoS, Delegated PoS, PBFT, and newer hybrid models.
- **Smart Contracts & Ethereum** – how programmable logic is stored on a blockchain.
- **Scalability Solutions** – sharding, layer 2 rollups, and sidechains.

- **Blockchain Security** – attacks (51 % attack, selfish mining) and mitigation strategies.

